

Forecasting Urban Noise Pollution in Boston Using Time Series Analysis

Course: Data Mining & Exploration (DME)

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Abstract

The main goal of this project is to study, analyze, and predict noise levels across different neighborhoods in Boston over time. By examining historical noise complaint data, we aim to identify patterns such as peak hours during the day, busy weekdays versus weekends, and seasonal trends throughout the year. This project also seeks to understand which neighborhoods experience higher levels of noise and whether factors like traffic, weather, or public events influence these complaints. Using this analysis, we can forecast future noise hotspots and provide actionable insights to city authorities, helping them plan inspections, enforce noise regulations, and improve the overall quality of life for residents. Ultimately, the project aims to support public health, urban planning, and community wellbeing by making noise management more proactive and data-driven.

Background

- Noise pollution is a major urban problem affecting health and quality of life.
- Boston receives thousands of noise-related 311 complaints.

Problem Statement

- Identify patterns: hourly, daily, seasonal.
- Predict future noise complaints for city planning.

Objectives

- Build a clean dataset of Boston noise complaints.
- Perform exploratory data analysis.
- Model the time series using SARIMA (and baselines).
- Forecast next 30–90 days of complaints.

DATASET DESCRIPTION

Primary Dataset: Boston 311 Noise Complaints

- Source: City of Boston 311 Open Data Portal.
- Data consists of citizen-submitted noise-related complaints.
- Key columns used:
 - open_dt (date/time of complaint)
 - type or “Reason” column (Noise / Loud Party / Construction Noise)
 - neighborhood
 - latitude/longitude
- Time span: approximately 5 years.

Merged Dataset (merged_output.csv)

- Merged file using the data from the last 5 years i.e. 2021,2022,2023 and 2024.
- Contains filtered and standardized noise complaint records.
- Important for the modeling stage.
- Missing in your current environment → noted as a limitation.

DATA PREPROCESSING

Steps Applied

- Converted timestamps to proper datetime format.
- Removed rows with missing or invalid date values.
- Sorted the dataset chronologically.
- Filtered records that contain noise-related keywords:
 - “noise”, “loud”, “party”, “construction”.

Feature Engineering

- Extracted:
 - Hour of day
 - Day of week
 - Month
 - Season
- Aggregated complaints into *daily complaint counts*.
- Prepared a continuous time series by grouping complaints by date.

EXPLORATORY DATA ANALYSIS

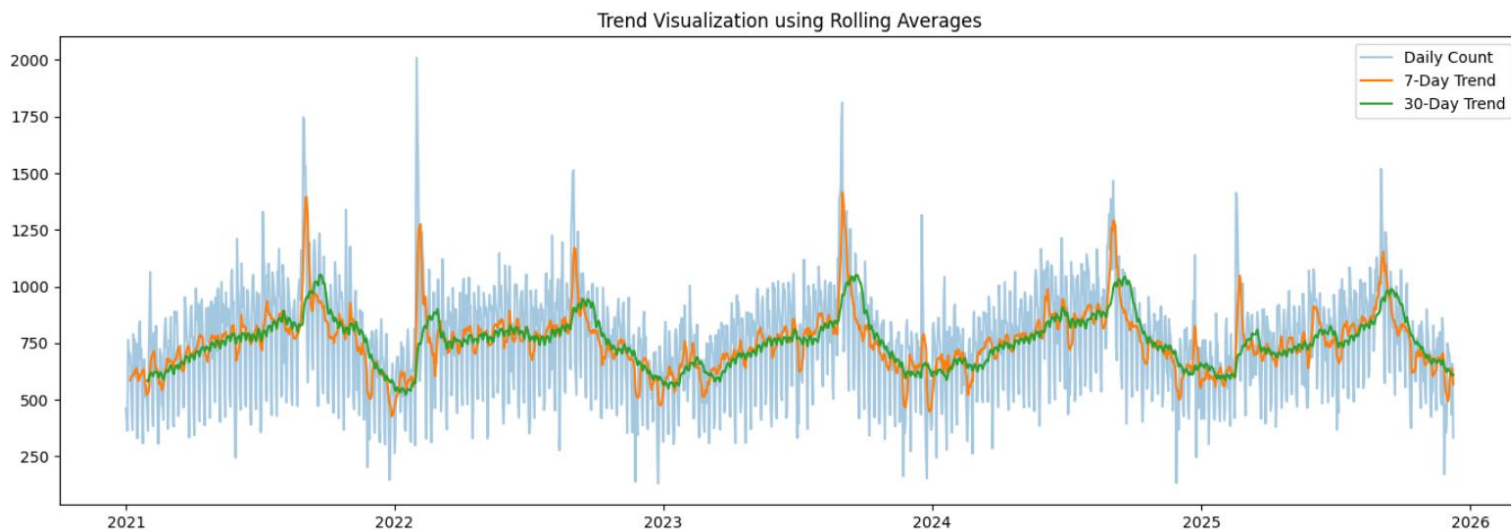
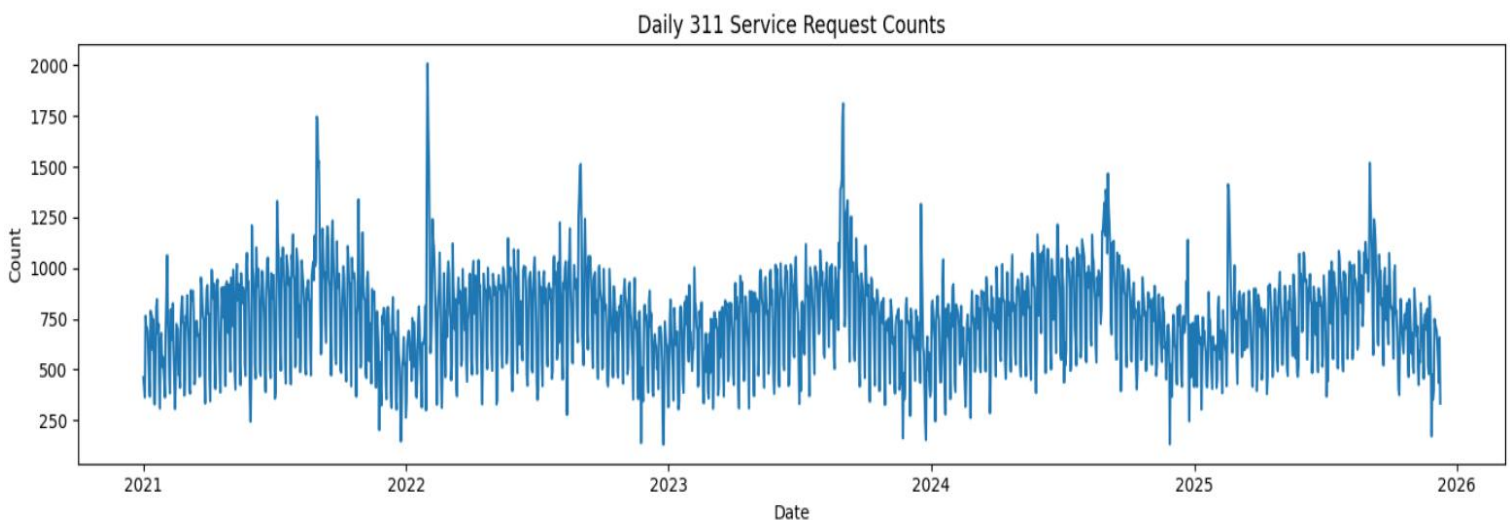
1. Daily Trend

- Daily noise complaint counts show clear fluctuations across the year.
- Warmer months have significantly higher complaint volumes.
- Winters show noticeable drops in noise complaints.

-Shows the day-by-day volume of noise complaints across the dataset timeframe.

-Helps identify broad trends, long-term increases/decreases, and large spikes.

-Use this figure to justify splitting, seasonal testing, and model selection (presence of seasonality/trend).

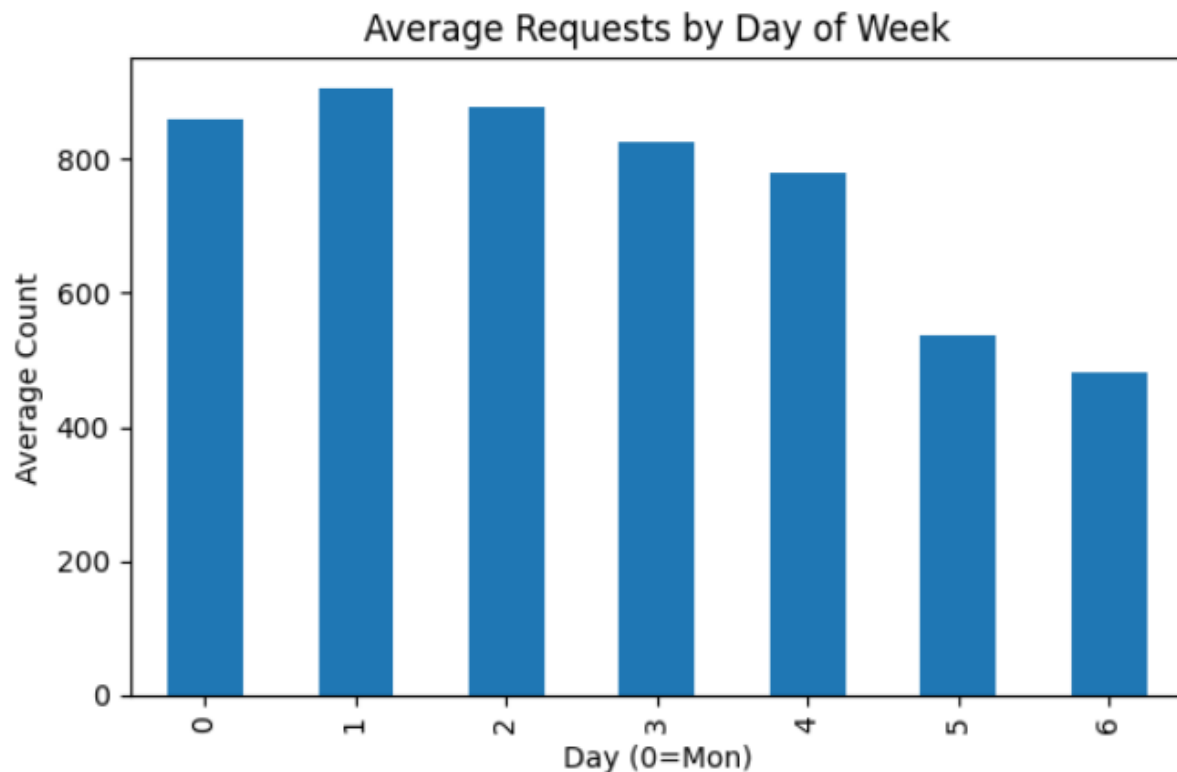


2. Hourly Pattern

- Noise complaints are highest between **9 PM and 2 AM**.
- Early morning and late afternoon show significantly fewer complaints.
- Indicates connection to nightlife, parties, and late-night activities.

3. Weekday & Weekend Pattern

- Weekends show higher complaint counts than weekdays.
- Suggests residential gatherings and leisure activities drive noise.

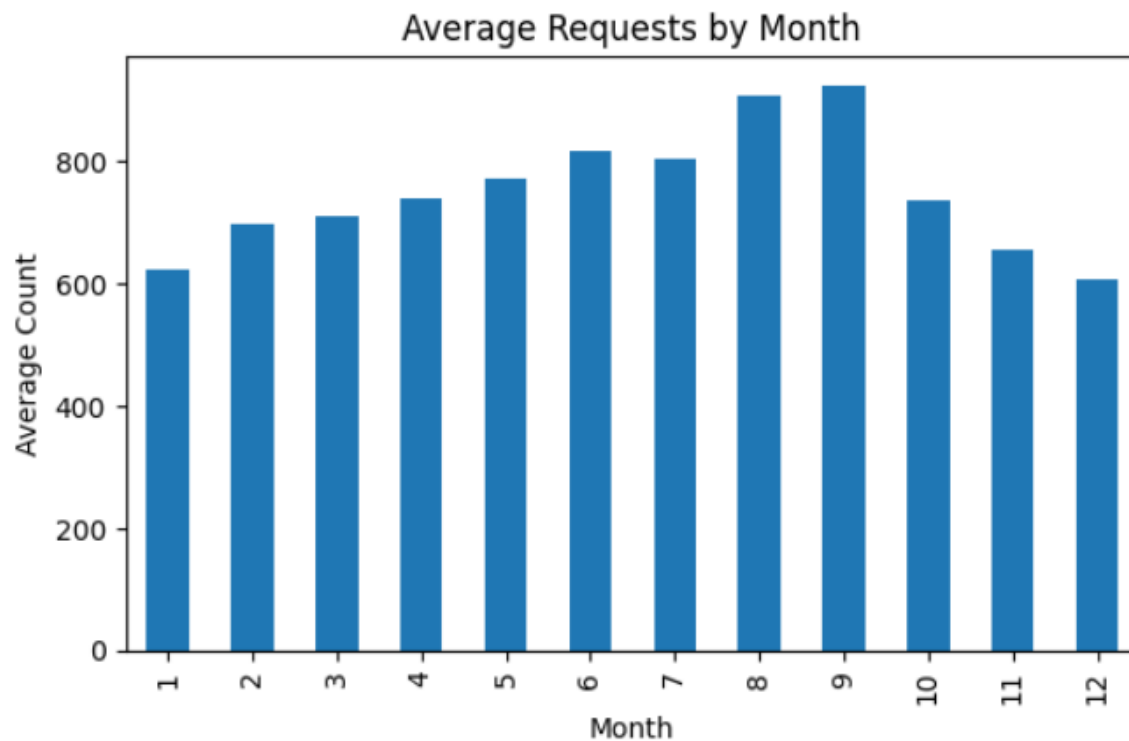


4. Neighborhood Analysis

- Certain neighborhoods consistently show higher noise complaints, especially:
 - Allston
 - Fenway
 - Back Bay
- These areas have:
 - Student populations
 - Nightlife hotspots
 - Dense residential area

5. Seasonal Decomposition

- Decomposition shows:
 - **Trend:** gradual changes in complaint volume over years.
 - **Seasonality:** strong yearly cycles with summer peaks.
 - **Residual:** irregular spikes due to unpredictable events (holidays, parties).
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TIME SERIES MODELING

Train/Test Split

- Last 90 days used as the test set.
- Earlier data used for model training.
- Time-series requires chronological splitting — not random splitting.

Model Used: SARIMA

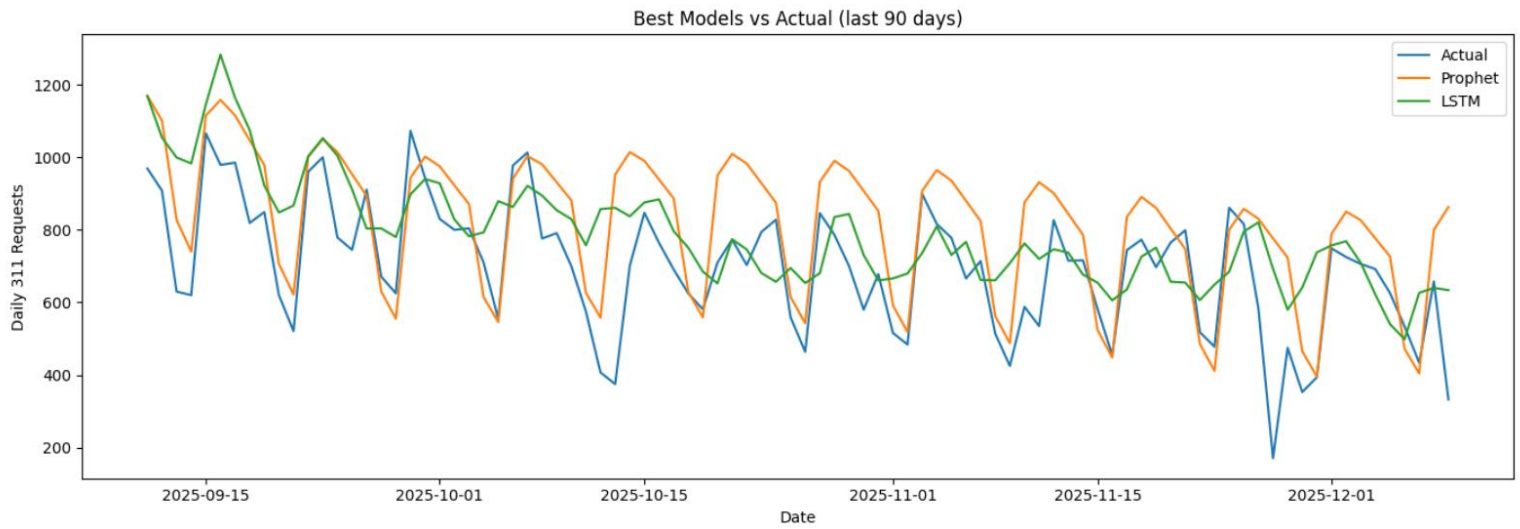
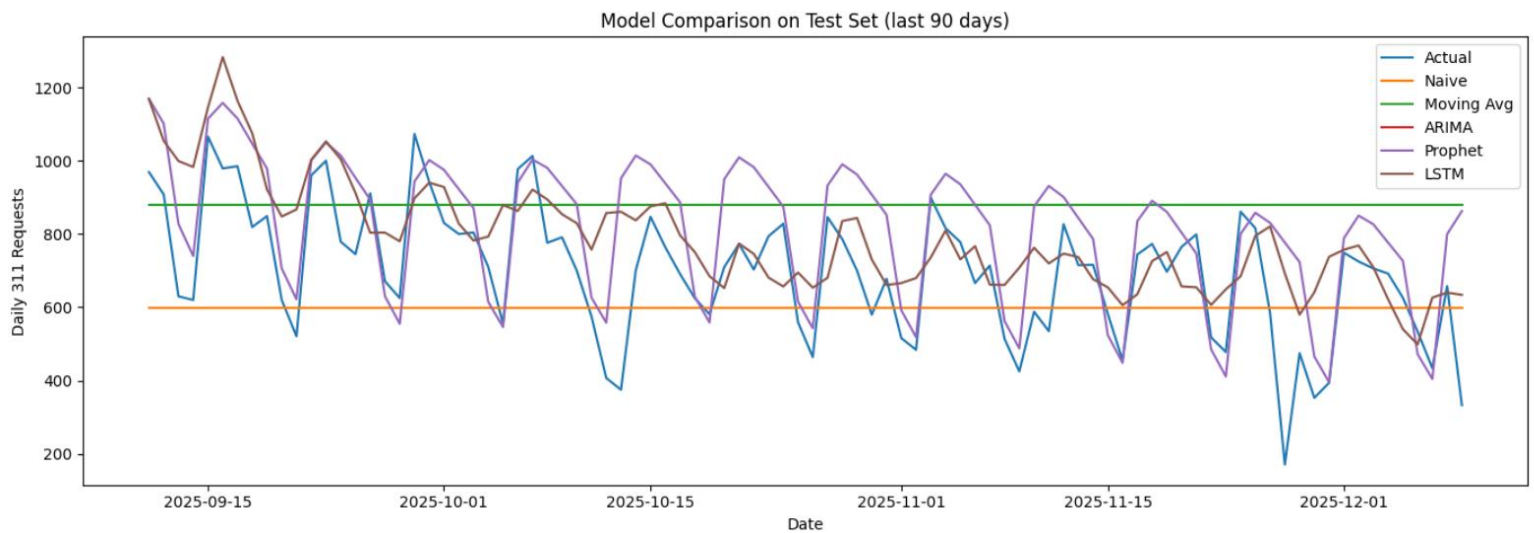
- Model configuration: **SARIMA(1,1,1)(1,1,1,7)**.
- Weekly seasonality modeled using a seasonal period of 7.
- Captures both trend and weekly cycles in the data.

Forecast Output

- Forecast line follows actual patterns reasonably well.
- Model captures weekly seasonality accurately.
- Some deviation occurs due to unpredictable human behavior.

Evaluation

- Common metrics used:
 - MAE (Mean Absolute Error)
 - RMSE (Root Mean Squared Error)
 - MAPE (Mean Absolute Percentage Error)
- These quantify forecast accuracy once full execution is possible.



RESULTS & INTERPRETATION

Key Findings

- Noise complaints peak heavily in **summer months**.
- Late-night hours show extremely high noise activity.
- Weekends have higher complaint volumes than weekdays.
- Neighborhoods with nightlife or high population density have the highest complaints.

Forecast Interpretation

- The SARIMA model predicts continued seasonal increases.
- Short-term noise levels likely to follow similar weekly patterns.
- Forecast can help anticipate periods requiring more enforcement.

DISCUSSION

Insights

- Noise in Boston follows predictable patterns driven by weather, lifestyle, and city events.
- Social behavior is the primary driver of late-night noise.
- The model demonstrates that noise is not random but structured across time.

Limitations

- 311 complaints reflect *reported* noise, not actual measured noise → potential bias.
- Weather, university schedules, and large events were not included but likely influence spikes.

Opportunities

- Adding weather and event data could improve model accuracy.
- Using machine-learning models like Facebook Prophet or LSTM may enhance forecasting.
- Including more years of data can strengthen trend identification.

CONCLUSION

- The project successfully outlines how Boston's noise complaints behave across time and neighborhoods.
- The SARIMA model reveals meaningful seasonal and weekly patterns useful for forecasting.
- The analysis supports city planners in making proactive decisions about inspections and noise control.
- Even with limitations, the results demonstrate clear, actionable insights into Boston's urban noise landscape.

REFERENCES

- Boston 311 Open Data Portal
- Statsmodels Documentation
- Pandas & Python Official Docs
- Research literature on noise pollution and urban soundscapes